

COLE A. MURRAY

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EDUCATION

University of Illinois at Urbana-Champaign, Grainger College of Engineering GPA: 3.76/4.00
Bachelor of Science in Computer Engineering, Hoeft Technology and Management Minor Expected Graduation May 2029

Related Coursework:

Digital Systems Laboratory, Analog Signal Processing, Computer Systems & Programming, Introduction to Computing, Introduction to Electronics, Linear Algebra with Computational Applications, Discrete Structures, Introduction to Differential Equations, Physics: Electricity and Magnetism, Physics: Mechanics

Awards: Grainger Engineering Visionary Scholarship, Schaumburg Business Association Scholarship Winner, Illinois State Seal of Biliteracy (Spanish)

Certifications: Anthropic: Claude Developer Platform 101, Intro to Model Context Protocol, Claude Code in Action

Involvements: Hoeft T&M Program, Phi Gamma Nu Professional Business Fraternity, Illinois Space Society, Illini Run Club

TECHNICAL SKILLS

Programming Languages: C/C++, Python, TypeScript, SystemVerilog, Java, SQL, ReactNative, JavaScript, Assembly (LC-3)

Frameworks/Tools: Microsoft Azure, Microsoft Entra, Git, Snowflake, Keboola, Supabase, MCP, Claude Code, Cursor, Vivado

WORK EXPERIENCE

Nerdio

Skokie, IL

Software Engineer Intern

May 2026 – August 2026

- Architected a **TypeScript MCP server** integrating Microsoft Graph and Azure ARM to audit privileged Entra ID / Azure RBAC access across a **60+-tenant multi-cloud identity platform**; 12 tools, OAuth app-only auth, and Azure Container Apps hosting with Entra-authenticated HTTP transport
- Built an authenticated **LLM/API proxy** so internal apps can call Claude without exposing the Anthropic API key; validates Entra ID JWTs, injects a server-side key, proxies the Anthropic API, and enforces per-user allowlists, model allowlists, and rate limits; deployed to Azure App Service with Key Vault secrets and structured audit logging
- Engineered a TypeScript **Azure Function App** multi-destination **webhook delivery platform** that replaced scattered automations with a queue-backed routing system to Teams, Azure Communication Services Email, or HTTPS webhooks, with an append-only Table Storage audit log; Entra Easy Auth and a custom RBAC admin dashboard on Static Web Apps, with GitHub Actions OIDC CI/CD

Solutions Engineer Intern

May 2025 – August 2025

- Designed and built internal tools to support product, customer success, and operations teams
- Developed interactive dashboards and apps to visualize product usage, customer interaction, and web traffic
- Created large scale data transformations to deliver actionable datasets to multiple departments
- Built and implemented multiple automation workflows, including a real time automated reply system for the company help forum and an email notification system for incoming support tickets
- Gained familiarity leveraging AI tools such as Cursor and Copilot 365 to increase and optimize my output

PERSONAL PROJECTS

First10 | Garmin Connect IQ, Monkey C, Embedded Programming, Constrained Hardware

- Built a Garmin wearable app that keeps an alarm active until a multi-sensor Awake Score (steps, sustained motion, and heart rate rise above a resting baseline) is passed, with sensors and alerts gated to the alarm phase
- Designed around Connect IQ limits: foreground-only vibration/tone and high-power CPU while open, keeping the AMOLED screen off and sensors idle during sleep, then enabling HR and accelerometer sampling when alarm fires
- Implemented difficulty presets, an HR-gated Hard mode resistant to wrist shakes, snooze caps, escalating alerts, and safety dismiss; added capability-checked wrappers, HR-unavailable fallback, and unit tests for scoring, hold timing, and degraded mode

Portfolio Website | TypeScript, React, Vercel

- Developed and deployed a responsive personal portfolio with TypeScript and React, showcasing projects and experience through a production site hosted on Vercel
- Created a powerful dev environment with 20+ plugins, MCPs, skills, and connectors to improve design taste